FUP - Oral Exam

neslusam and question providers

Contents

Introduction 1
Functional Programming 1
Racket 1
λ-Calculus 1
Haskell 2

⚠️ **TRIGGER WARNING - M*NAD** ⚠️ - The text below contains repeated, explicit occurences of the "M-Word".If you are currently recovering from trying to understand what a m*nad actually is, further review of this material is not recommended..

Introduction

[V] - Tomáš Votroubek , [H] - Rostislav Horčík , [Z] - Matěj Zorek **Your examiner is random, your previous lecturer doesn't make a difference.**

There might be a newer version of this document, check out: [FUP Oral Exam GitHub Repository]()

Functional Programming

[Z] What is pure/Higher-order function + examples of each possible combination?

[Z] List types of recursions based on branching factor + examples/classify head/tail recursion.

Racket

[V] How to implement infinite stream of nats/ones

1. *using stream functions?*
2. *without using stream functions?*

[V] Describe how does foldl/foldr work, which parametres are used and what is the difference between foldl and foldr.

λ-Calculus

[Z] List and describe terms in syntax of λ-calculus.

[H/Z] What is a redex?

[H] What is a reduction? List and describe different types of reductions and their usage. (left right inner outer TODO:Remove)

[V] What is a normal form and how to transform a λ-expression to its normal form?

[Z] What is the difference between bound and free variable? Which operation is done on them?

[H] What is Church-Rosser theorem?

[H] how to represent True/False in λ-Calculus?

[H] How to represent number in λ-Calculus? How to represent addition of 2 numbers?

[H] reduce this expression $(\lambda x.x(\lambda x.xx))e$ to its normal form.

[V] Find all free/bound vars , redexes in (λ f. (λ x. f (x x)) (λ x. f (x x))). Find normal form.

Haskell

[V] Describe signature of function map.

[V] Describe signature of function foldl/r.

[H] What is a typeclass?

[H] How to implement polymorphic typeclass , which adds 1 to an Int and prints out length for String?

[V] What is Functor typeclass? Which functions define it?

[V] What is a Applicative typeclass? Which functions define it?

[V] What is a Monad? Which functions define it?

[H] What is a State Monad, what is it good for? How to implement bind for State Monad?

You are all done!